

How well does the
regression line fit
the data?

Fitting

How well does the
regression line fit
the data?

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regression line fit
the data?



Accuracy of
prediction

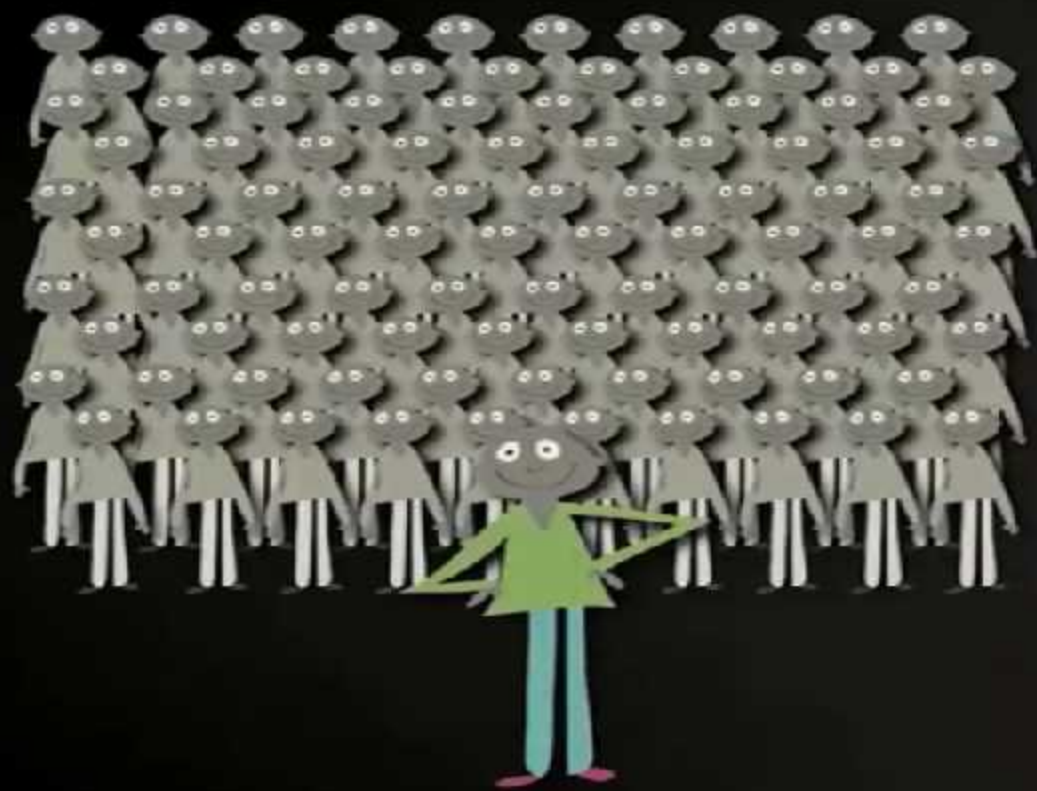
How well does the
regression line fit
the data?



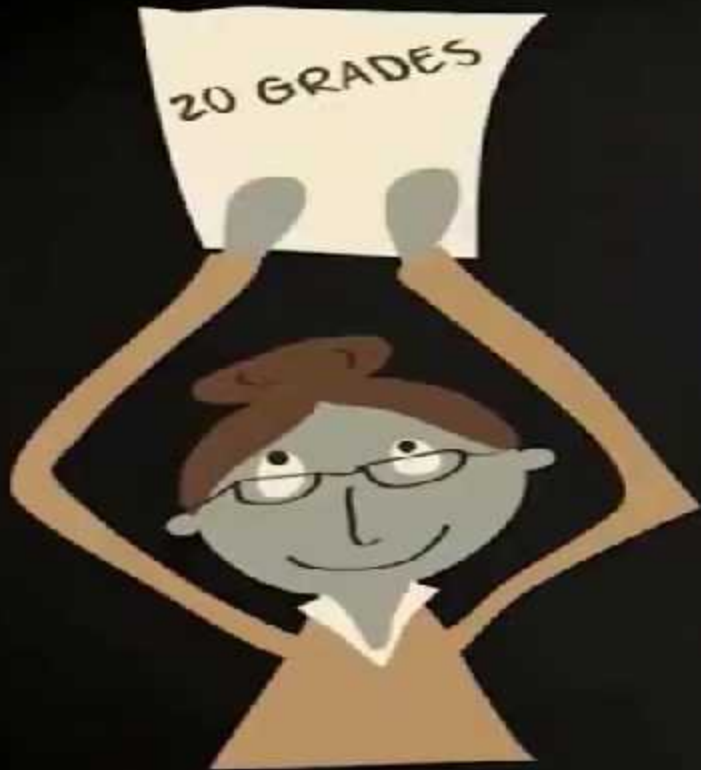
Accuracy of
prediction



$$r^2$$







grade exam

student 1	3.1
student 2	3.4
student 3	4.4
student 4	5.1
student 5	5.7
student 6	6
student 7	6.5
student 8	6.7
student 9	6.9
student 10	7.1
student 11	7.2
student 12	7.3
student 13	7.5
student 14	7.8
student 15	8
student 16	8.2
student 17	8.6
student 18	8.7
student 19	9.5
student 20	9.8

grade exam

student 1	3.1
student 2	3.4
student 3	4.4
student 4	5.1
student 5	5.7
student 6	6
student 7	6.5
student 8	6.7
student 9	6.9
student 10	7.1
student 11	7.2
student 12	7.3
student 13	7.5
student 14	7.8
student 15	8
student 16	8.2
student 17	8.6
student 18	8.7
student 19	9.5
student 20	9.8

mean = 6.8

	grade exam	grade prev. exam
student 1	3.1	5
student 2	3.4	4
student 3	4.4	6
student 4	5.1	4
student 5	5.7	6.2
student 6	6	7
student 7	6.5	7
student 8	6.7	6.2
student 9	6.9	7.4
student 10	7.1	7.4
student 11	7.2	7.9
student 12	7.3	6.9
student 13	7.5	7.7
student 14	7.8	6.9
student 15	8	7.5
student 16	8.2	8
student 17	8.6	7.1
student 18	8.7	9
student 19	9.5	7.6
student 20	9.8	8

mean = 6.8

	grade exam	grade prev. exam
student 1	3.1	5
student 2	3.4	4
student 3	4.4	6
student 4	5.1	4
student 5	5.7	6.2
student 6	6	7
student 7	6.5	7
student 8	6.7	6.2
student 9	6.9	7.4
student 10	7.1	7.4
student 11	7.2	7.9
student 12	7.3	6.9
student 13	7.5	7.7
student 14	7.8	6.9
student 15	8	7.5
student 16	8.2	8
student 17	8.6	7.1
student 18	8.7	9
student 19	9.5	7.6
student 20	9.8	8

mean = 6.8



regression analysis

$$\hat{y} = 2.80 + 0.59x$$



$$\hat{y} = 2.80 + 0.59x$$



$$x = 8.1$$

$$\hat{y} = 2.80 + 0.59x$$



$$x = 8.1 \quad \hat{y} = 2.80 + (0.59 * 8.1)$$

$$\hat{y} = 2.80 + 0.59x$$

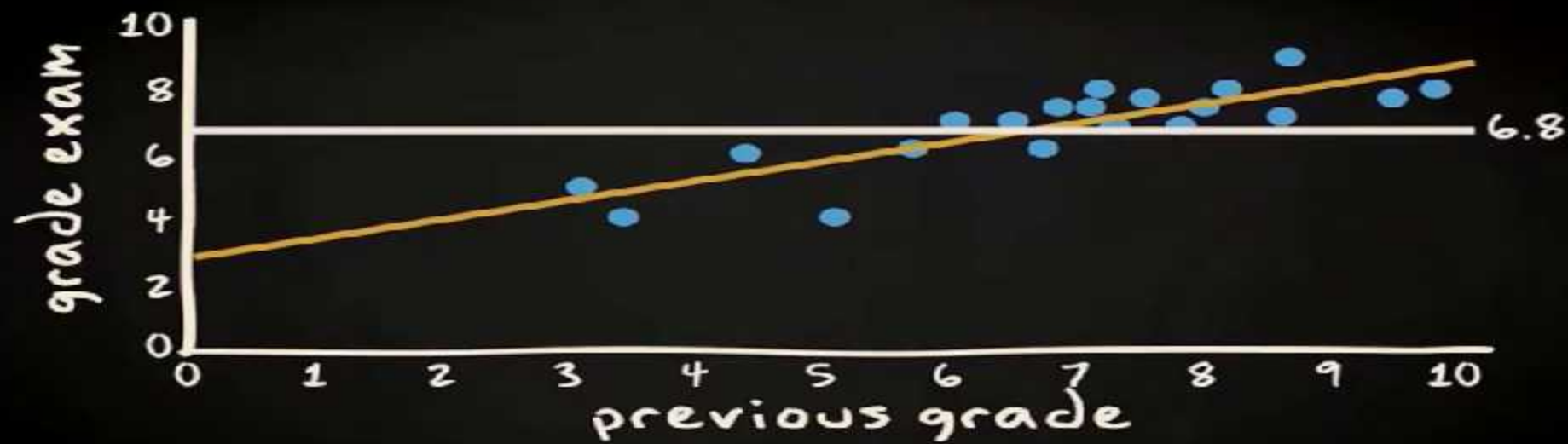


$$x = 8.1 \quad \hat{y} = 2.80 + (0.59 * 8.1) = 7.6$$

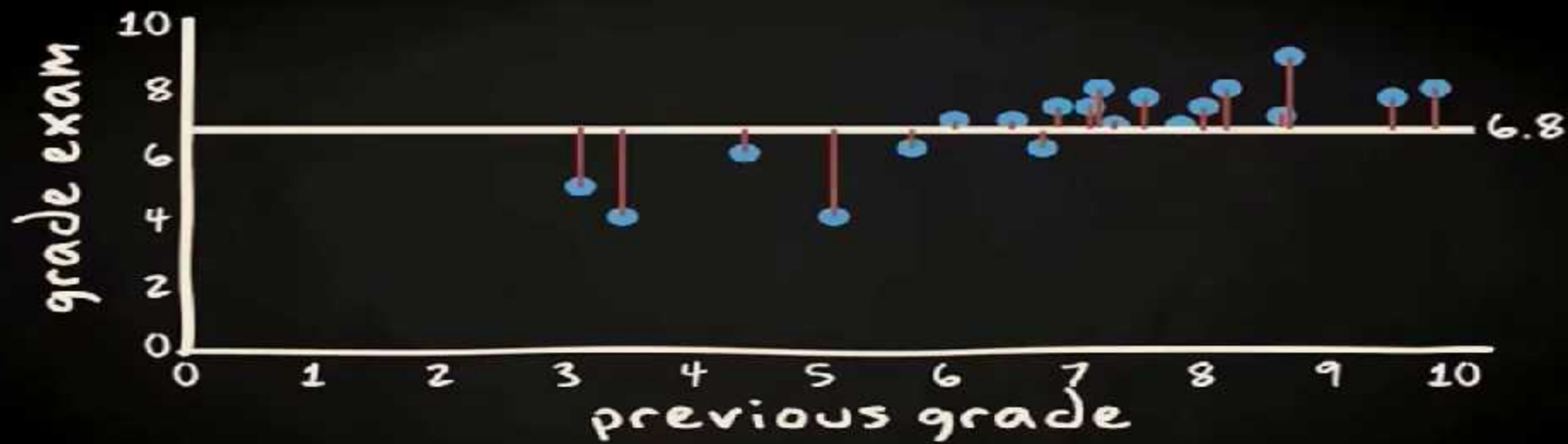
r^2

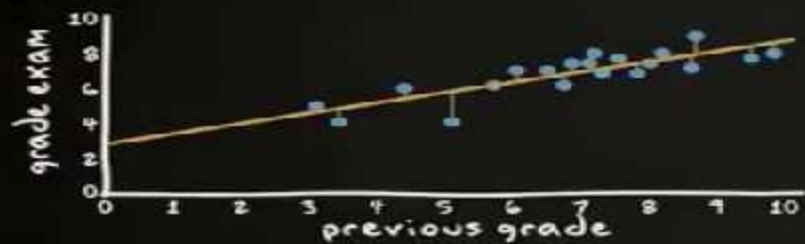


tells you how much
better a
regression line predicts
the value of a
dependent variable
than the mean of the
variable









regression line



mean

better
prediction

$$r^2 = 0.69$$

$$r^2 = 0.69$$



prediction error
is 69% smaller
than when you
use the mean

$$r^2 = 0.69$$

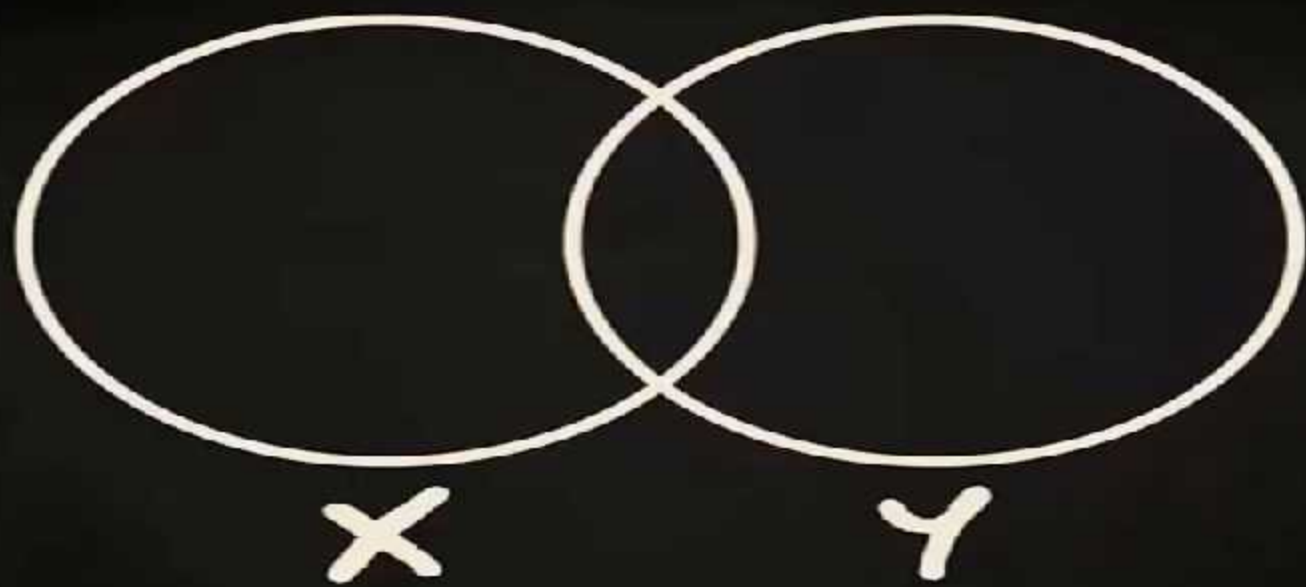


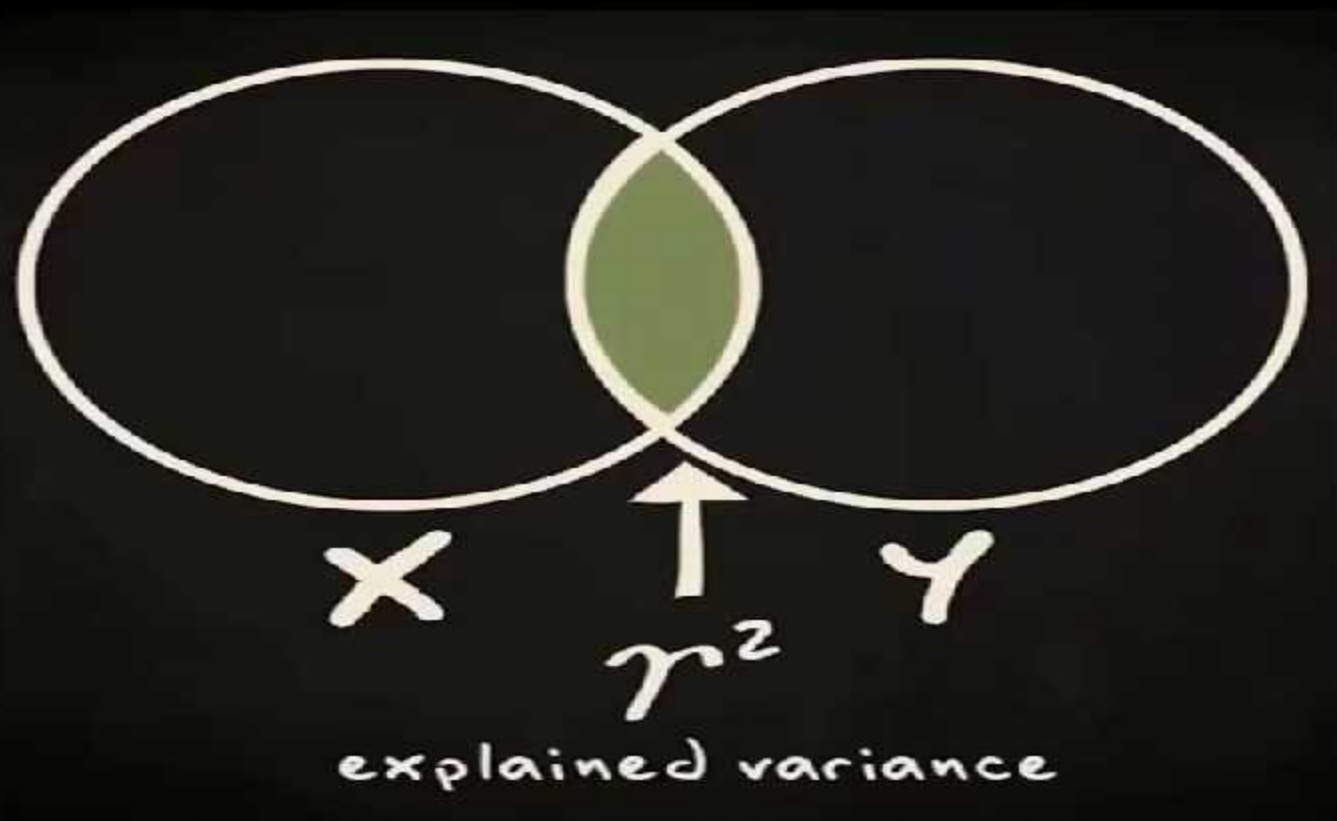
the amount of
variance in your
dependent variable
(Y) that is explained
by your independent
variable (X)

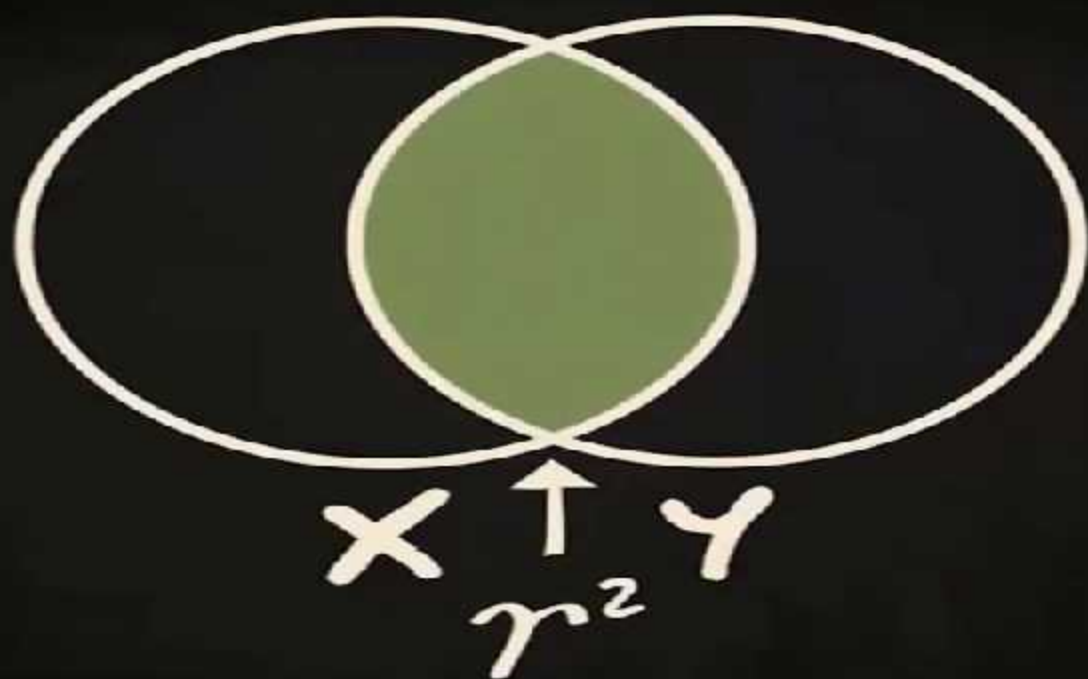
$$r^2 = 0.69$$



69% of the
variance in the
grades can be
predicted by the
previous grades







explained variance

Pearson's r



r^2

Pearson's r



square



r^2

Pearson's r



square



r^2

(always positive)

$$r = 0.83$$



square



$$r^2$$

(always positive)

$$r = 0.83$$



square



$$r^2 = 0.69$$

$$r = 1$$



Square



$$r^2 = 1$$

$$r = 0$$



square



$$r^2 = 0$$

r

direction and
strength of relation

r

direction and
strength of relation

 r^2

r

direction and
strength of relation

r^2

1. how much better a
regression line predicts your
dependent variable than the
mean of that variable

r

direction and
strength of relation

 r^2

1. how much better a regression line predicts your dependent variable than the mean of that variable
2. how much of the variance in your dependent variable is explained by your independent variable

x

$x \rightarrow y$

$X \rightarrow Y$







r^2 = how well does the
line fit the data



CAREFUL INTERPRETATION

1. correlation
 $\neq$
 causation
- 2.

chocolate
consumption (x)

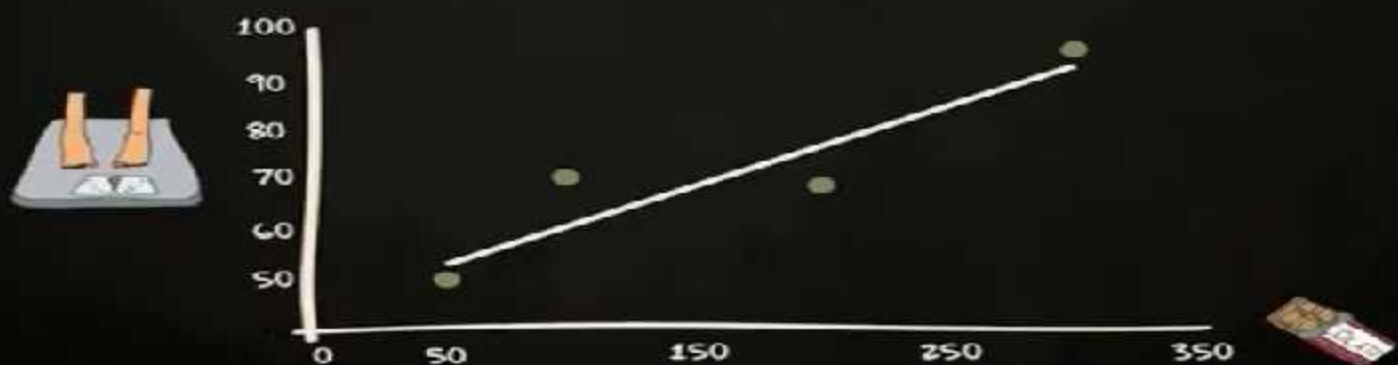


body weight
(y)

chocolate
consumption (x)



body weight
(y)



chocolate
consumption (x)

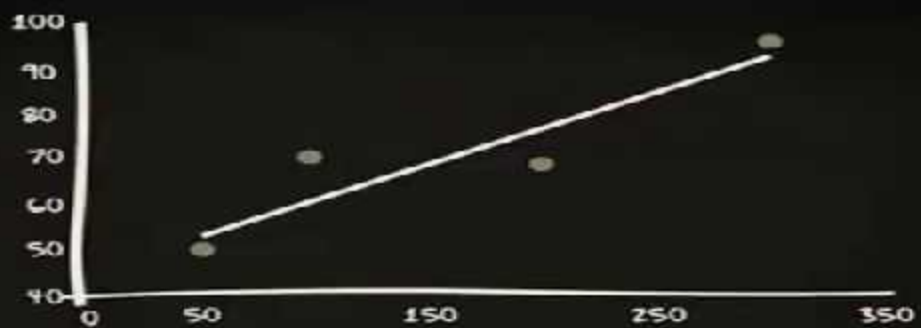


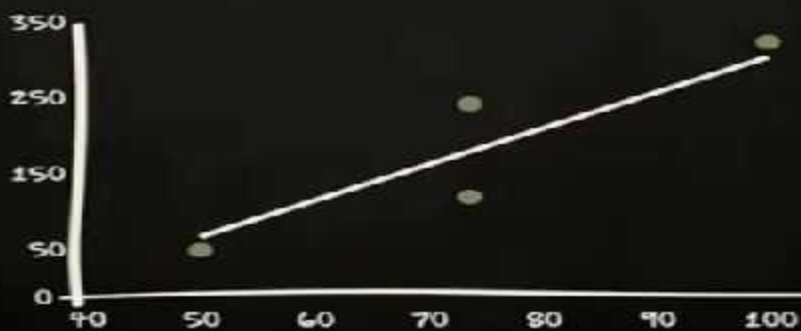
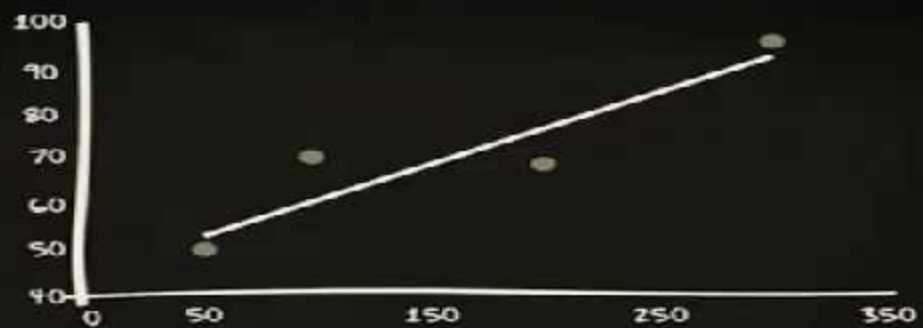
body weight
(y)

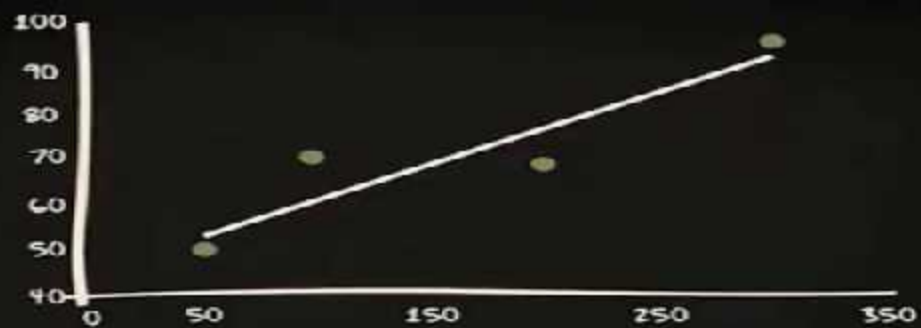
body weight
(x)



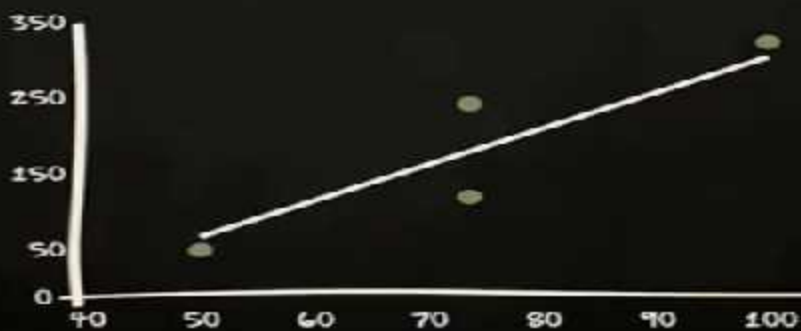
chocolate
consumption (y)







$$\hat{y} = 46.15 + 0.15x$$



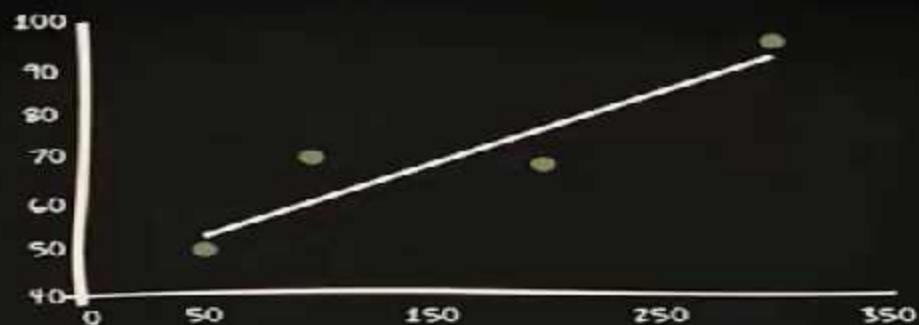
$$\hat{y} = -235.28 + 5.58x + 0.15x^2$$



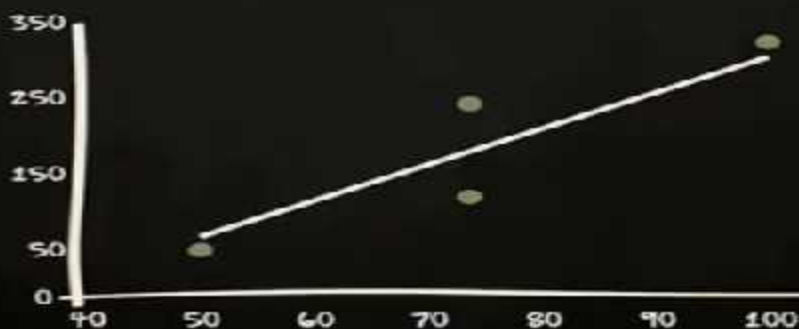
chocolate
consumption



body weight



$$\hat{y} = 46.15 + 0.15x$$



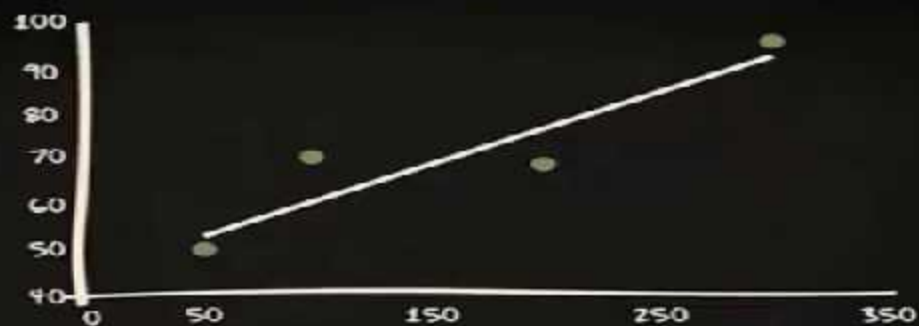
$$\hat{y} = -235.28 + 5.58x + 0.15x^2$$



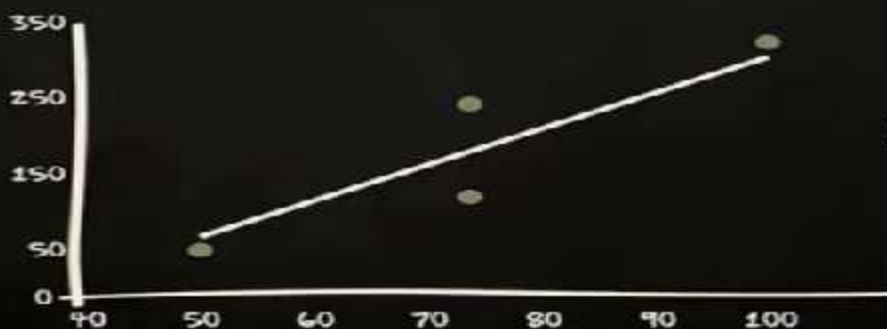
chocolate
consumption



body weight



$$\hat{y} = 46.15 + 0.15x$$



$$\hat{y} = -235.28 + 5.58x + 0.15x^2$$

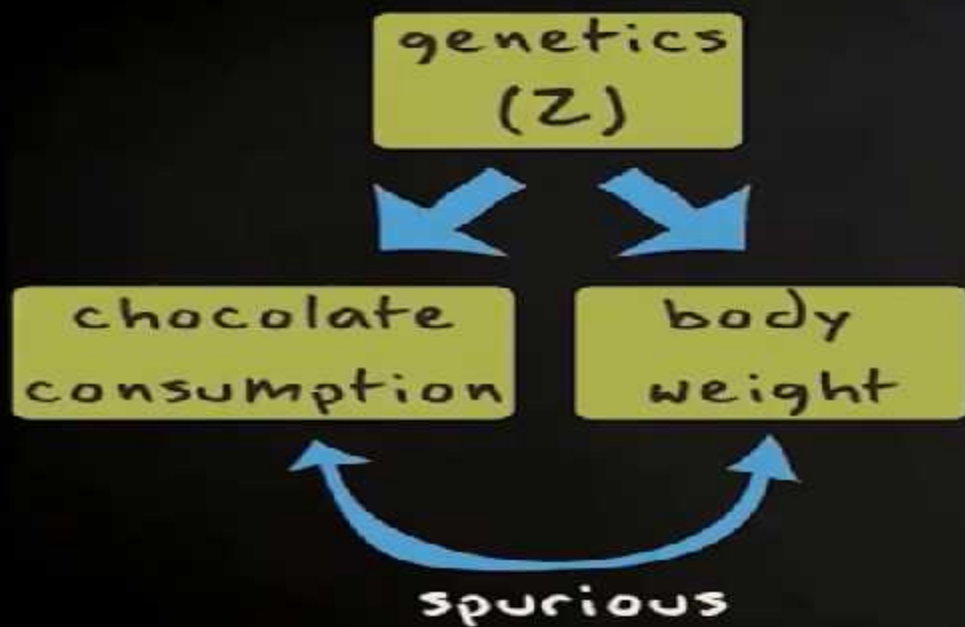


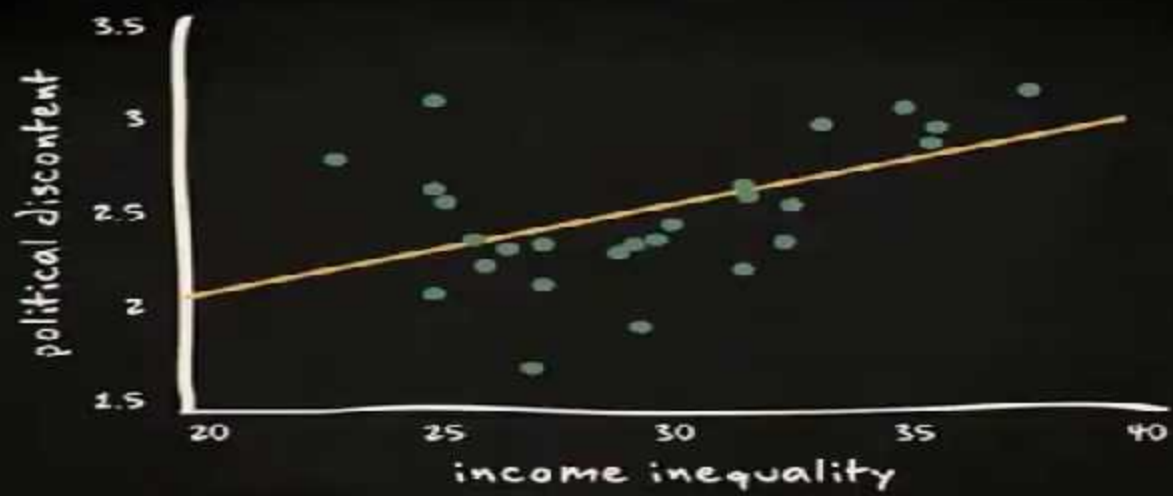
genetics
(Z)

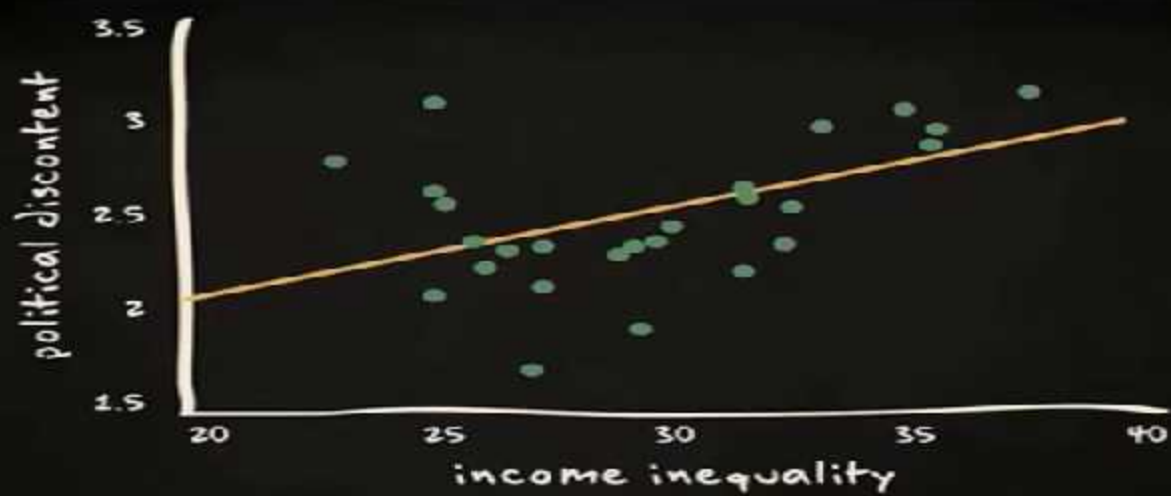


chocolate
consumption

body
weight

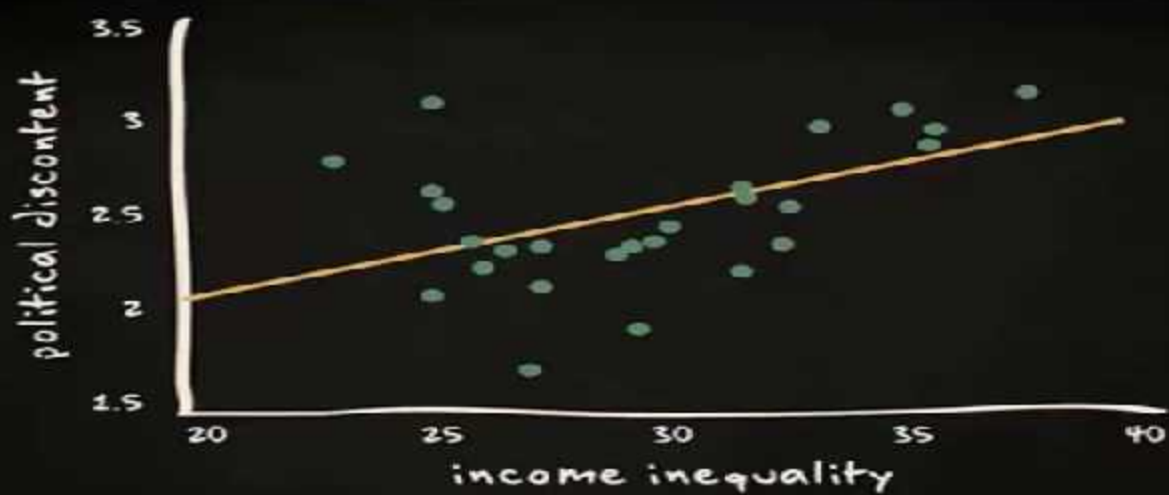






income
inequality





income
inequality



political
discontent

corruption



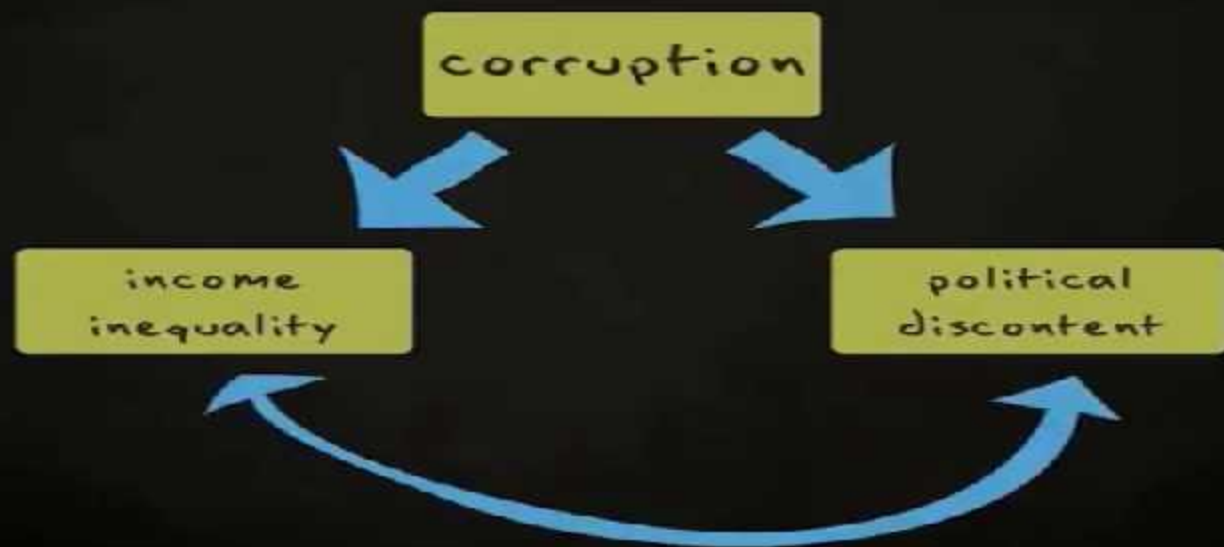
income
inequality



political
discontent



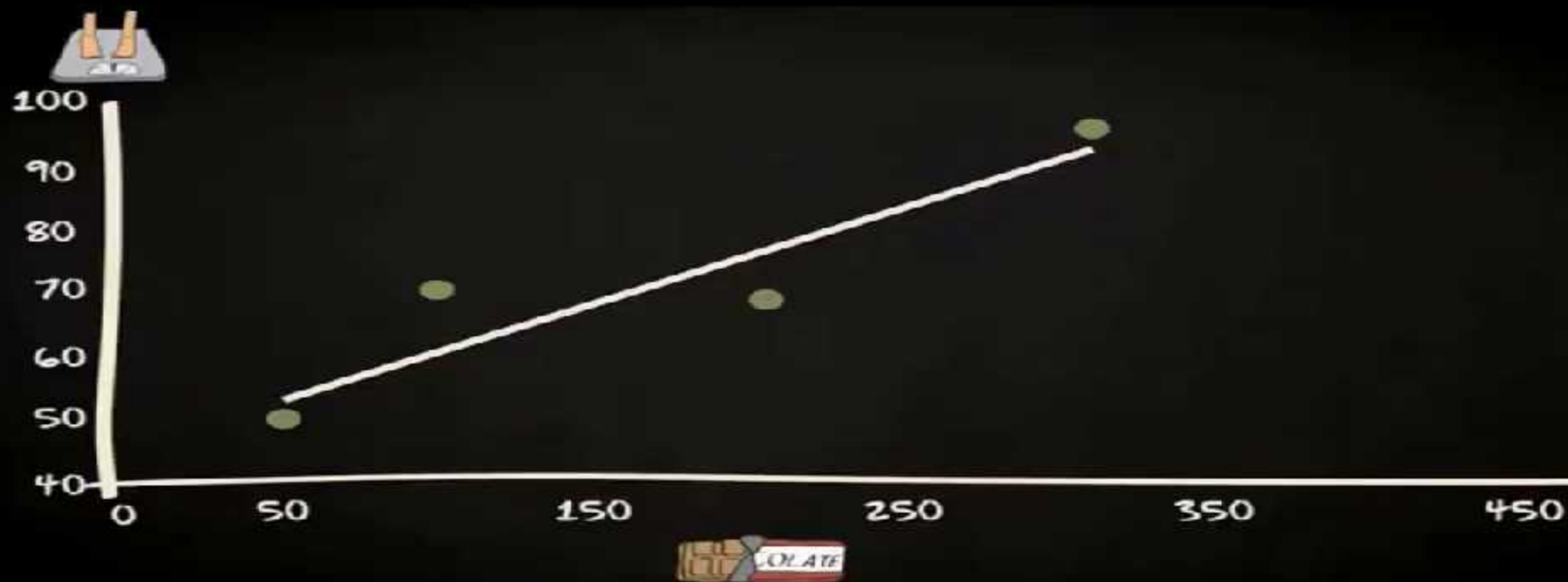


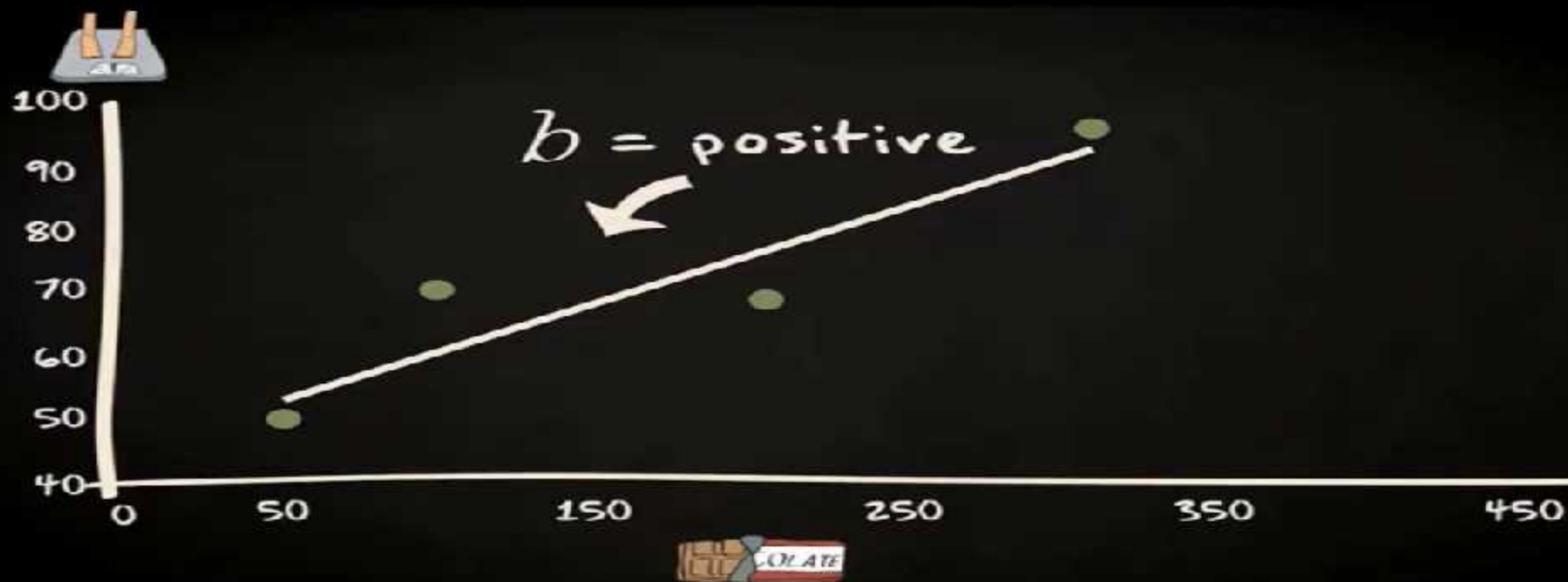




CAREFUL INTERPRETATION

1. correlation
 $\neq$
 causation
2. be aware
 of
 outliers



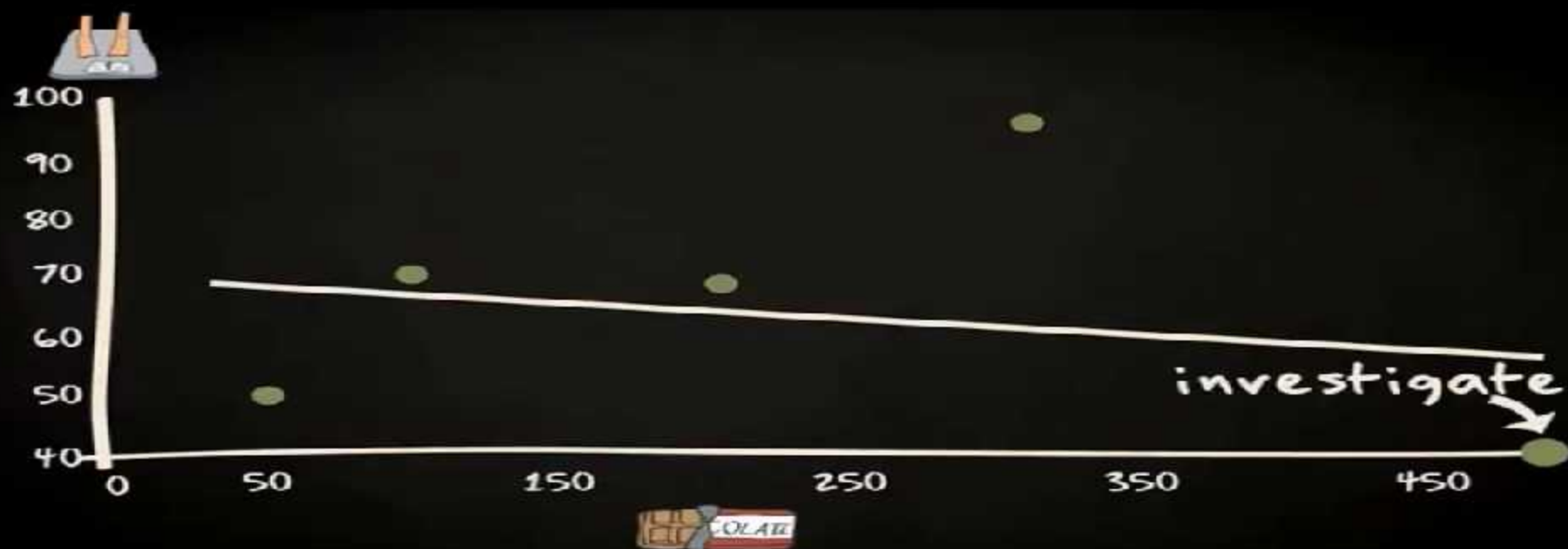
















LOW



LOW





X 261





X 261



survey

Survey

Extraterrestrial life
is highly plausible

Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Correlated?



Correlated?



level of
measurement?



Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Correlated?

level of
measurement?
↓
ordinal



Survey

Extraterrestrial life is highly plausible

- ☐ Agree
- ☐ Neither agree nor disagree
- ☐ Disagree

Level of education

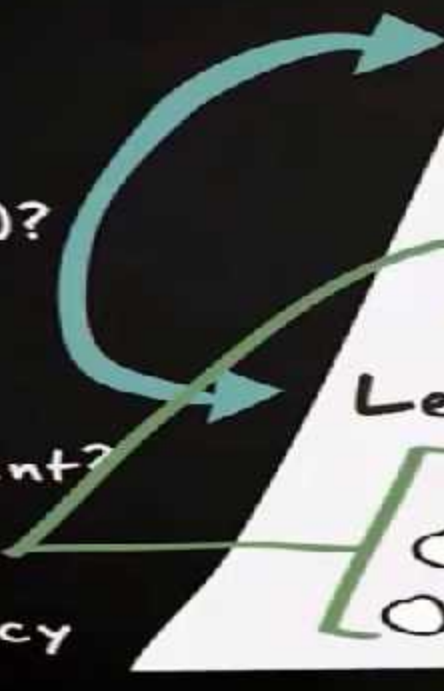
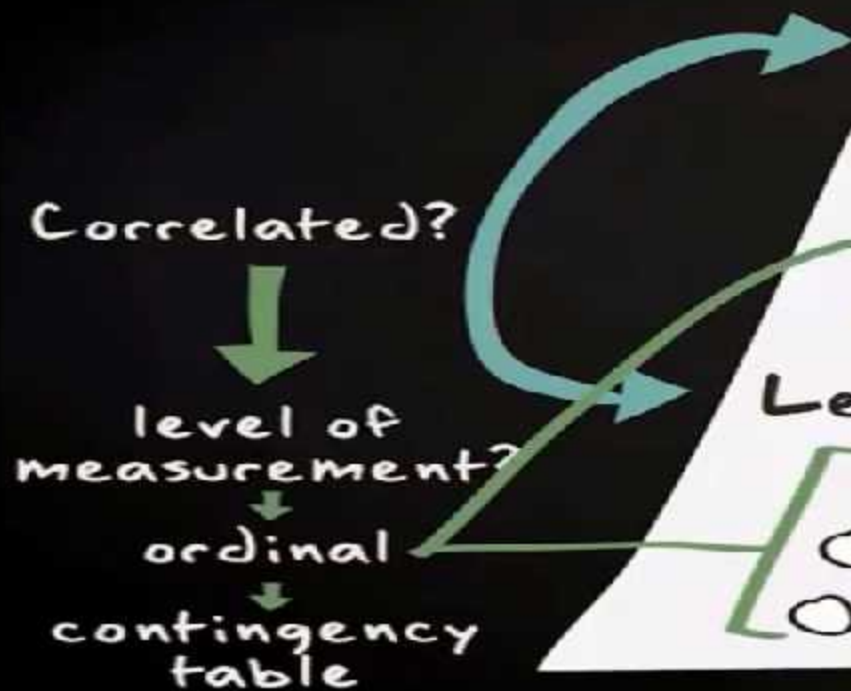
- ☐ High
- ☐ Middle
- ☐ Low

Correlated?

level of measurement?

ordinal

contingency table



Survey

Extraterrestrial life
is highly plausible

- ☐ Agree
- ☐ Neither agree
nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Correlated?

level of
measurement?
↓
ordinal
↓
contingency
table

scatterplot
Pearson R
OLS regression

Survey

Extraterrestrial life is highly plausible

- ☐ Agree
- ☐ Neither agree nor disagree
- ☐ Disagree

Level of education

- ☐ High
- ☐ Middle
- ☐ Low

Correlated?

level of measurement?
↓
ordinal
↓
contingency table

level of measurement?

↓
quantitative

↓
scatterplot
Pearson R
OLS regression

contingency table



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



IV



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



IV



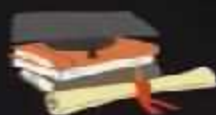
	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261

DV



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261

DV



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261

Correlation?



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261

Correlation?

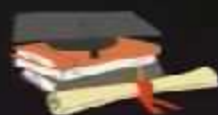
relevant percentages



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
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total	50	109	102	261

Correlation?

column percentages



	High	Middle	Low	total
Agree	25	31	12	68
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column percentages



	High	Middle	Low	total
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column percentages



*100% =



	High	Middle	Low	total
Agree	25	31	12	68
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Disagree	8	40	64	112
total	50	109	102	261



column percentages

25

* 100% =



	High	Middle	Low	total
Agree	25	31	12	68
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Disagree	8	40	64	112
total	50	109	102	261



column percentages



25
50

*100% =



	High	Middle	Low	total
Agree	25	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages

25
50

*100% = 50%



	High	Middle	Low	total
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Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages

25
50

*100% = 50%



	High	Middle	Low	total
Agree	50%	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages

26

* 100% =



	High	Middle	Low	total
Agree	50%	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages



26
102

* 100% =



	High	Middle	Low	total
Agree	50%	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages

$$\frac{26}{102} \times 100\% = 25\%$$



	High	Middle	Low	total
Agree	50%	31	12	68
Agree/disagree	17	38	26	81
Disagree	8	40	64	112
total	50	109	102	261



column percentages

$$\frac{26}{102} * 100\% = 25\%$$



	High	Middle	Low	total
Agree	50%	31	12	68
Agree/disagree	17	38	25%	81
Disagree	8	40	64	112
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column percentages



	High	Middle	Low	total
Agree	50%	28%	12%	26%
Agree/disagree	34%	35%	25%	31%
Disagree	16%	37%	63%	43%
total	100%	100%	100%	100%



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	High	Middle	Low	total
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